A* Algorithm

AI-10.py - C:/Users/Vineela/OneDrive/Documents/AI-10.py (3.15.0)

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```
import heapq

graph = {
    "A": [("B", 1), ("C", 4)],
    "B": [("D", 2), ("E", 5)],
    "C": [("E", 1)],
    "D": [("G", 5)],
    "E": [("G", 3)],
    "G": []
}

h = {
    "A": 7,
    "B": 6,
    "C": 4,
    "D": 4,
    "E": 2,
    "G": 0
}

def astar(start, goal):
    queue = [(h[start], 0, start, [start])]
    best_cost = {start: 0}

    while queue:
        f, cost, node, path = heapq.heappop(queue)

        if cost != best_cost.get(node):
            continue

        if node == goal:
            print("Path:", path)
            print("Total cost:", cost)
            return

        for neighbor, weight in graph[node]:
            new_cost = cost + weight

            if new_cost < best_cost.get(neighbor, float("inf")):
                best_cost[neighbor] = new_cost
                priority = new_cost + h[neighbor]

                heapq.heappush(
                    queue,
                    (priority, new_cost, neighbor, path + [neighbor])
                )

    print("No path found.")

astar("A", "G")
```

IDLE Shell 3.15.0

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```
Python 3.15.0 (tags/v3.15.0:d220409, Oct  9 2026, 11:55:56) [MSC v.1951 64 bit (
AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
============= RESTART: C:/Users/Vineela/OneDrive/Documents/AI-10.py ============
Path: ['A', 'B', 'D', 'G']
Total cost: 8
>>>
```